A6B2_EXPT-8: BLACK BOX TESTING

Test Case Document - Online Leave Management System (Website)

Particular	Details
Course	Software Engineering Lab (24CS01PR0502)
Experiment	8 - Black Box Testing
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Class / Batch	5S CSE / A6 - B2
Roll No.	28
Date of test execution	25-10-2026
Application under test	Online Leave Management System - faculty leave portal (4 form interfaces)
Faculty	D A Borikar

Aim

To compile a test case document for a website having at least three form interfaces (excluding Login/Register forms) in a prescribed template, indicating the inputs and the outputs of every test case.

Problem Statement / Application Under Test

The Online Leave Management System lets faculty members submit leave requests. The system validates and processes each request, supports approval or rejection by the administrator, maintains faculty and leave records, and sends status notifications by Email, SMS or Mobile App. The website under test has four form interfaces, none of which is a Login or Register form:

- Form 1 - Apply for Leave (apply-leave.html): A faculty member submits a Casual (CL), Medical (ML) or On-Duty (OD) leave request and chooses how to be notified.
- Form 2 - Update Faculty Profile (profile.html): A faculty member updates the contact details used for leave notifications.
- Form 3 - Approve / Reject Leave (approve-leave.html): The administrator approves or rejects a pending leave request; the faculty member is notified on the preferred channel.
- Form 4 - Leave Balance Enquiry (leave-balance.html): Shows the available and total leave of a faculty member for an academic year.

Black Box Testing - Techniques Used

Black box testing checks the software against its specification without looking at the internal code. The tester supplies inputs through the user interface and compares the output shown with the expected output.

- Equivalence Partitioning (EP): each input is divided into valid and invalid classes, and one representative value from each class is tested.
- Boundary Value Analysis (BVA): values at and just outside the limits (min-1, min, max, max+1) are tested, because errors are most likely at the edges of a range.

Test Environment

Item	Details
Website	Online Leave Management System (HTML, CSS, JavaScript), served locally at http://127.0.0.1:8008
Browser	Microsoft Edge 156.0.0.0 (headless) on Windows 11
Execution	Each test case was executed through the user interface: the page is opened, the input data is entered in the fields, the submit button is clicked and the message displayed on the page is recorded as the actual output (website/tests/run-tests.html).
Executed on	Sun Oct 25 2026 19:17:54 GMT+0530 (India Standard Time)

Test Data (Pre-loaded Records)

Table 1 - Faculty records and leave balance (2026-27), shown as available / total days

Faculty ID	Name	Category	Dept.	Channel	CL	ML	OD
FAC101	Dr. Anita Deshmukh	Permanent	CSE	Email	8 / 12	10 / 10	5 / 7
FAC102	Mr. Rahul Patil	Contract	CSE	SMS	5 / 8	Not eligible	3 / 5
FAC103	Dr. Sneha Kulkarni	Permanent	IT	Mobile App	12 / 12	2 / 10	7 / 7

Table 2 - Leave requests

Request ID	Faculty ID	Leave type	From	Days	Status
LR-1001	FAC101	CL	02-11-2026	2	PENDING
LR-1002	FAC102	OD	05-11-2026	3	PENDING
LR-1003	FAC103	ML	20-10-2026	1	APPROVED

All data is reset when a page is loaded, so every test case starts from the state above.

Form Interfaces - Inputs and Outputs

Form 1: Apply for Leave (apply-leave.html)

Input field	Control	Valid input	Output (message) for invalid input
Faculty ID	Text box	FAC + 3 digits; must be registered (case-insensitive, spaces trimmed)	Faculty ID is required. / Faculty ID must be FAC followed by 3 digits (e.g. FAC101). / Faculty ID <id> is not registered.
Leave type	Drop-down	CL, ML or OD	Please select a leave type.
From date	Date	Today or a later date	From date is required. / From date cannot be in the past.
To date	Date	Same as or after From date	To date is required. / To date cannot be before From date.
Number of days	Read-only (output)	Calculated as To date - From date + 1	-
Reason	Text area	10 to 200 characters	Reason must be 10 to 200 characters.
Notify me by	Radio buttons	Email, SMS or Mobile App	Please select a notification channel.
Leave policy (checked after all fields are valid)	Business rule	Contract faculty cannot take ML; max per request CL 3, ML 15, OD 10 days; days must not exceed available balance	Contract faculty are not eligible for Medical Leave. / <Leave name> cannot exceed <n> days per request. / Insufficient <type> balance (available: <n> day(s)).

Output for valid input: Leave request LR-2001 submitted successfully (<type>, <n> day(s), status PENDING, notification via <channel>).

Form 2: Update Faculty Profile (profile.html)

Input field	Control	Valid input	Output (message) for invalid input
Faculty ID	Text box	FAC + 3 digits; must be registered	Same three messages as Form 1.
Full name	Text box	3 to 50 characters; letters, spaces and dots only	Name must be 3 to 50 characters (letters, spaces and dots only).
Email	Text box	name@domain.tld format	Enter a valid email address (e.g. name@college.edu).
Mobile number	Text box	Exactly 10 digits, first digit 6, 7, 8 or 9	Mobile number must be 10 digits starting with 6, 7, 8 or 9.
Department	Drop-down	CSE, IT, ECE, EE, ME or CE	Please select a department.

Output for valid input: Profile updated successfully for <id> (<name>, <dept>).

Form 3: Approve / Reject Leave (approve-leave.html)

Input field	Control	Valid input	Output (message) for invalid input
Request ID	Text box	LR- + 4 digits; must exist and be PENDING	Request ID is required. / Request ID must be LR- followed by 4 digits (e.g. LR-1001). / Leave request <id> not found. / Leave request <id> is already <status> and cannot be changed.
Decision	Radio buttons	Approve or Reject	Please select Approve or Reject.
Remarks	Text area	Max 200 characters; at least 10 characters when rejecting	Remarks cannot exceed 200 characters. / Remarks of at least 10 characters are required when rejecting.

Output for valid input: Leave request <id> APPROVED|REJECTED. Status notification sent to <faculty ID> via <channel>.

Form 4: Leave Balance Enquiry (leave-balance.html)

Input field	Control	Valid input	Output (message) for invalid input
Faculty ID	Text box	FAC + 3 digits; must be registered	Same three messages as Form 1.
Academic year	Drop-down	2026-27 or 2025-26	Please select an academic year.
Leave type	Drop-down	All (default), CL, ML or OD	-

Output for valid input: Leave balance for <id> (<year>): CL a/t, ML a/t, OD a/t (a = available, t = total) and a balance table; for a year without records: No leave records found for <id> in <year>.

Test Case Document

Template: TC ID | Test scenario (technique) | Pre-condition | Test steps | Input (test data) | Expected output | Actual output | Status. Dates are shown as dd-mm-yyyy; the date of execution was 25-10-2026.

Form 1: Apply for Leave

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_AL_01	Valid Casual Leave request of 2 days [EP (valid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 26-10-2026 To date: 27-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	Leave request LR-2001 submitted successfully (CL, 2 day(s), status PENDING, notification via Email).	Leave request LR-2001 submitted successfully (CL, 2 day(s), status PENDING, notification via Email).	**PASS**
TC_AL_02	Faculty ID left blank [EP (invalid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: (blank) Leave type: CL From date: 26-10-2026 To date: 27-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	Faculty ID is required.	Faculty ID is required.	**PASS**
TC_AL_03	Faculty ID in wrong format (only 2 digits) [EP (invalid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC12 Leave type: CL From date: 26-10-2026 To date: 27-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	Faculty ID must be FAC followed by 3 digits (e.g. FAC101).	Faculty ID must be FAC followed by 3 digits (e.g. FAC101).	**PASS**
TC_AL_04	Faculty ID in correct format but not registered [EP (invalid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC999 Leave type: CL From date: 26-10-2026 To date: 27-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	Faculty ID FAC999 is not registered.	Faculty ID FAC999 is not registered.	**PASS**
TC_AL_05	From date one day in the past [BVA (today - 1)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 24-10-2026 To date: 26-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	From date cannot be in the past.	From date cannot be in the past.	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_AL_06	Leave for today only (From = To = today) [BVA (today)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 25-10-2026 To date: 25-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	Leave request LR-2001 submitted successfully (CL, 1 day(s), status PENDING, notification via Email).	Leave request LR-2001 submitted successfully (CL, 1 day(s), status PENDING, notification via Email).	**PASS**
TC_AL_07	To date earlier than From date [EP (invalid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 28-10-2026 To date: 27-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	To date cannot be before From date.	To date cannot be before From date.	**PASS**
TC_AL_08	Casual Leave of exactly 3 days (upper limit) [BVA (max = 3)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 26-10-2026 To date: 28-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	Leave request LR-2001 submitted successfully (CL, 3 day(s), status PENDING, notification via Email).	Leave request LR-2001 submitted successfully (CL, 3 day(s), status PENDING, notification via Email).	**PASS**
TC_AL_09	Casual Leave of 4 days (above limit) [BVA (max + 1)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 26-10-2026 To date: 29-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Email	Casual Leave cannot exceed 3 days per request.	Casual Leave cannot exceed 3 days per request.	**PASS**
TC_AL_10	Contract faculty applies for Medical Leave [EP (invalid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC102 Leave type: ML From date: 26-10-2026 To date: 27-10-2026 Reason: "Attending a family function" (27 chars) Notify by: SMS	Contract faculty are not eligible for Medical Leave.	Contract faculty are not eligible for Medical Leave.	**PASS**
TC_AL_11	Medical Leave of 3 days with only 2 days balance [EP (invalid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC103 Leave type: ML From date: 26-10-2026 To date: 28-10-2026 Reason: "Attending a family function" (27 chars) Notify by: Mobile App	Insufficient ML balance (available: 2 day(s)).	Insufficient ML balance (available: 2 day(s)).	**PASS**
TC_AL_12	Reason of 9 characters [BVA (min - 1)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 26-10-2026 To date: 27-10-2026 Reason: "Need rest" (9 chars) Notify by: Email	Reason must be 10 to 200 characters.	Reason must be 10 to 200 characters.	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_AL_13	Reason of exactly 10 characters [BVA (min)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 26-10-2026 To date: 27-10-2026 Reason: "Need rest." (10 chars) Notify by: Email	Leave request LR-2001 submitted successfully (CL, 2 day(s), status PENDING, notification via Email).	Leave request LR-2001 submitted successfully (CL, 2 day(s), status PENDING, notification via Email).	**PASS**
TC_AL_14	No notification channel selected [EP (invalid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: CL From date: 26-10-2026 To date: 27-10-2026 Reason: "Attending a family function" (27 chars) Notify by: (not selected)	Please select a notification channel.	Please select a notification channel.	**PASS**
TC_AL_15	Valid On-Duty Leave of 5 days, Mobile App notification [EP (valid)]	Apply for Leave page open; pre-loaded faculty FAC101, FAC102, FAC103 (Table 1).	1. Open Apply for Leave page 2. Enter input data 3. Click 'Submit request'	Faculty ID: FAC101 Leave type: OD From date: 30-10-2026 To date: 03-11-2026 Reason: "Conference at IIT Bombay" (24 chars) Notify by: Mobile App	Leave request LR-2001 submitted successfully (OD, 5 day(s), status PENDING, notification via Mobile App).	Leave request LR-2001 submitted successfully (OD, 5 day(s), status PENDING, notification via Mobile App).	**PASS**

Form 2: Update Faculty Profile

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_FP_01	All profile fields valid [EP (valid)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Dr. Anita Deshmukh" (18 chars) Email: anita.deshmukh@college.edu Mobile: 9876543210 Department: CSE	Profile updated successfully for FAC101 (Dr. Anita Deshmukh, CSE).	Profile updated successfully for FAC101 (Dr. Anita Deshmukh, CSE).	**PASS**
TC_FP_02	Name contains digits [EP (invalid)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Anita123" (8 chars) Email: anita.deshmukh@college.edu Mobile: 9876543210 Department: CSE	Name must be 3 to 50 characters (letters, spaces and dots only).	Name must be 3 to 50 characters (letters, spaces and dots only).	**PASS**
TC_FP_03	Name of 2 characters [BVA (min - 1)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Al" (2 chars) Email: anita.deshmukh@college.edu Mobile: 9876543210 Department: CSE	Name must be 3 to 50 characters (letters, spaces and dots only).	Name must be 3 to 50 characters (letters, spaces and dots only).	**PASS**
TC_FP_04	Name of exactly 3 characters [BVA (min)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Ali" (3 chars) Email: anita.deshmukh@college.edu Mobile: 9876543210 Department: CSE	Profile updated successfully for FAC101 (Ali, CSE).	Profile updated successfully for FAC101 (Ali, CSE).	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_FP_05	Name of exactly 50 characters [BVA (max)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Dr. Anita Deshmukh Ramchandra Kulkarni Patil Joshi" (50 chars) Email: anita.deshmukh@college.edu Mobile: 9876543210 Department: CSE	Profile updated successfully for FAC101 (Dr. Anita Deshmukh Ramchandra Kulkarni Patil Joshi, CSE).	Profile updated successfully for FAC101 (Dr. Anita Deshmukh Ramchandra Kulkarni Patil Joshi, CSE).	**PASS**
TC_FP_06	Name of 51 characters [BVA (max + 1)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Dr. Anita Deshmukh Ramchandra Kulkarni Patil Joshii" (51 chars) Email: anita.deshmukh@college.edu Mobile: 9876543210 Department: CSE	Name must be 3 to 50 characters (letters, spaces and dots only).	Name must be 3 to 50 characters (letters, spaces and dots only).	**PASS**
TC_FP_07	Email without @ symbol [EP (invalid)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Dr. Anita Deshmukh" (18 chars) Email: anita.deshmukh.college.edu Mobile: 9876543210 Department: CSE	Enter a valid email address (e.g. name@college.edu).	Enter a valid email address (e.g. name@college.edu).	**PASS**
TC_FP_08	Mobile number of 9 digits [BVA (length - 1)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Dr. Anita Deshmukh" (18 chars) Email: anita.deshmukh@college.edu Mobile: 987654321 Department: CSE	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	**PASS**
TC_FP_09	Mobile number of 11 digits [BVA (length + 1)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Dr. Anita Deshmukh" (18 chars) Email: anita.deshmukh@college.edu Mobile: 98765432101 Department: CSE	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	**PASS**
TC_FP_10	Mobile number starting with 5 [EP (invalid)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Dr. Anita Deshmukh" (18 chars) Email: anita.deshmukh@college.edu Mobile: 5876543210 Department: CSE	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	**PASS**
TC_FP_11	Department not selected [EP (invalid)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC101 Full name: "Dr. Anita Deshmukh" (18 chars) Email: anita.deshmukh@college.edu Mobile: 9876543210 Department: (not selected)	Please select a department.	Please select a department.	**PASS**
TC_FP_12	Faculty ID not registered [EP (invalid)]	Update Profile page open; faculty FAC101 registered.	1. Open Update Faculty Profile page 2. Enter input data 3. Click 'Save profile'	Faculty ID: FAC999 Full name: "Dr. Anita Deshmukh" (18 chars) Email: anita.deshmukh@college.edu Mobile: 9876543210 Department: CSE	Faculty ID FAC999 is not registered.	Faculty ID FAC999 is not registered.	**PASS**

Form 3: Approve / Reject Leave

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_AP_01	Approve a pending request without remarks [EP (valid)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1001 Decision: Approve Remarks: (blank)	Leave request LR-1001 APPROVED. Status notification sent to FAC101 via Email.	Leave request LR-1001 APPROVED. Status notification sent to FAC101 via Email.	**PASS**
TC_AP_02	Reject a pending request with remarks [EP (valid)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1002 Decision: Reject Remarks: "Insufficient staff on duty" (26 chars)	Leave request LR-1002 REJECTED. Status notification sent to FAC102 via SMS.	Leave request LR-1002 REJECTED. Status notification sent to FAC102 via SMS.	**PASS**
TC_AP_03	Reject without remarks [EP (invalid)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1002 Decision: Reject Remarks: (blank)	Remarks of at least 10 characters are required when rejecting.	Remarks of at least 10 characters are required when rejecting.	**PASS**
TC_AP_04	Reject with 9-character remarks [BVA (min - 1)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1002 Decision: Reject Remarks: "Not valid" (9 chars)	Remarks of at least 10 characters are required when rejecting.	Remarks of at least 10 characters are required when rejecting.	**PASS**
TC_AP_05	Reject with exactly 10-character remarks [BVA (min)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1002 Decision: Reject Remarks: "Not valid." (10 chars)	Leave request LR-1002 REJECTED. Status notification sent to FAC102 via SMS.	Leave request LR-1002 REJECTED. Status notification sent to FAC102 via SMS.	**PASS**
TC_AP_06	Request ID left blank [EP (invalid)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: (blank) Decision: Approve Remarks: (blank)	Request ID is required.	Request ID is required.	**PASS**
TC_AP_07	Request ID without LR- prefix [EP (invalid)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: 1001 Decision: Approve Remarks: (blank)	Request ID must be LR- followed by 4 digits (e.g. LR-1001).	Request ID must be LR- followed by 4 digits (e.g. LR-1001).	**PASS**
TC_AP_08	Request ID that does not exist [EP (invalid)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-9999 Decision: Approve Remarks: (blank)	Leave request LR-9999 not found.	Leave request LR-9999 not found.	**PASS**
TC_AP_09	Request already approved earlier [EP (invalid)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1003 Decision: Approve Remarks: (blank)	Leave request LR-1003 is already APPROVED and cannot be changed.	Leave request LR-1003 is already APPROVED and cannot be changed.	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_AP_10	Decision not selected [EP (invalid)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1001 Decision: (not selected) Remarks: (blank)	Please select Approve or Reject.	Please select Approve or Reject.	**PASS**
TC_AP_11	Approve with remarks of exactly 200 characters [BVA (max)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1001 Decision: Approve Remarks: 'A' x 200 (200 characters)	Leave request LR-1001 APPROVED. Status notification sent to FAC101 via Email.	Leave request LR-1001 APPROVED. Status notification sent to FAC101 via Email.	**PASS**
TC_AP_12	Approve with remarks of 201 characters [BVA (max + 1)]	Approve / Reject page open; LR-1001 and LR-1002 PENDING, LR-1003 already APPROVED (Table 2).	1. Open Approve / Reject Leave page 2. Enter input data 3. Click 'Record decision'	Request ID: LR-1001 Decision: Approve Remarks: 'A' x 201 (201 characters)	Remarks cannot exceed 200 characters.	Remarks cannot exceed 200 characters.	**PASS**

Form 4: Leave Balance Enquiry

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_LB_01	Permanent faculty, all leave types [EP (valid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: FAC101 Academic year: 2026-27 Leave type: All	Leave balance for FAC101 (2026-27): CL 8/12, ML 10/10, OD 5/7	Leave balance for FAC101 (2026-27): CL 8/12, ML 10/10, OD 5/7	**PASS**
TC_LB_02	Permanent faculty, Casual Leave only [EP (valid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: FAC101 Academic year: 2026-27 Leave type: CL	Leave balance for FAC101 (2026-27): CL 8/12	Leave balance for FAC101 (2026-27): CL 8/12	**PASS**
TC_LB_03	Contract faculty, all leave types (ML not eligible) [EP (valid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: FAC102 Academic year: 2026-27 Leave type: All	Leave balance for FAC102 (2026-27): CL 5/8, ML Not eligible, OD 3/5	Leave balance for FAC102 (2026-27): CL 5/8, ML Not eligible, OD 3/5	**PASS**
TC_LB_04	Permanent faculty, Medical Leave only [EP (valid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: FAC103 Academic year: 2026-27 Leave type: ML	Leave balance for FAC103 (2026-27): ML 2/10	Leave balance for FAC103 (2026-27): ML 2/10	**PASS**
TC_LB_05	Academic year with no records [EP (valid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: FAC101 Academic year: 2025-26 Leave type: All	No leave records found for FAC101 in 2025-26.	No leave records found for FAC101 in 2025-26.	**PASS**
TC_LB_06	Faculty ID in lower case with spaces [EP (valid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: " fac101 " (with spaces) Academic year: 2026-27 Leave type: All	Leave balance for FAC101 (2026-27): CL 8/12, ML 10/10, OD 5/7	Leave balance for FAC101 (2026-27): CL 8/12, ML 10/10, OD 5/7	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_LB_07	Faculty ID left blank [EP (invalid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: (blank) Academic year: 2026-27 Leave type: All	Faculty ID is required.	Faculty ID is required.	**PASS**
TC_LB_08	Faculty ID with letter O instead of zero [EP (invalid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: FAC1O1 Academic year: 2026-27 Leave type: All	Faculty ID must be FAC followed by 3 digits (e.g. FAC101).	Faculty ID must be FAC followed by 3 digits (e.g. FAC101).	**PASS**
TC_LB_09	Faculty ID not registered [EP (invalid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: FAC999 Academic year: 2026-27 Leave type: All	Faculty ID FAC999 is not registered.	Faculty ID FAC999 is not registered.	**PASS**
TC_LB_10	Academic year not selected [EP (invalid)]	Leave Balance page open; balances as in Table 1 (records exist only for 2026-27).	1. Open Leave Balance Enquiry page 2. Enter input data 3. Click 'Show balance'	Faculty ID: FAC101 Academic year: (not selected) Leave type: All	Please select an academic year.	Please select an academic year.	**PASS**

Test Execution Summary

Form	Test cases executed	Passed	Failed
Form 1: Apply for Leave	15	15	0
Form 2: Update Faculty Profile	12	12	0
Form 3: Approve / Reject Leave	12	12	0
Form 4: Leave Balance Enquiry	10	10	0
Total	**49**	**49**	**0**

Execution Screenshots

Screenshots were captured in Microsoft Edge after each test case was executed. The dark bar at the top is added by the test harness and shows the test case ID, the time of execution and the result.

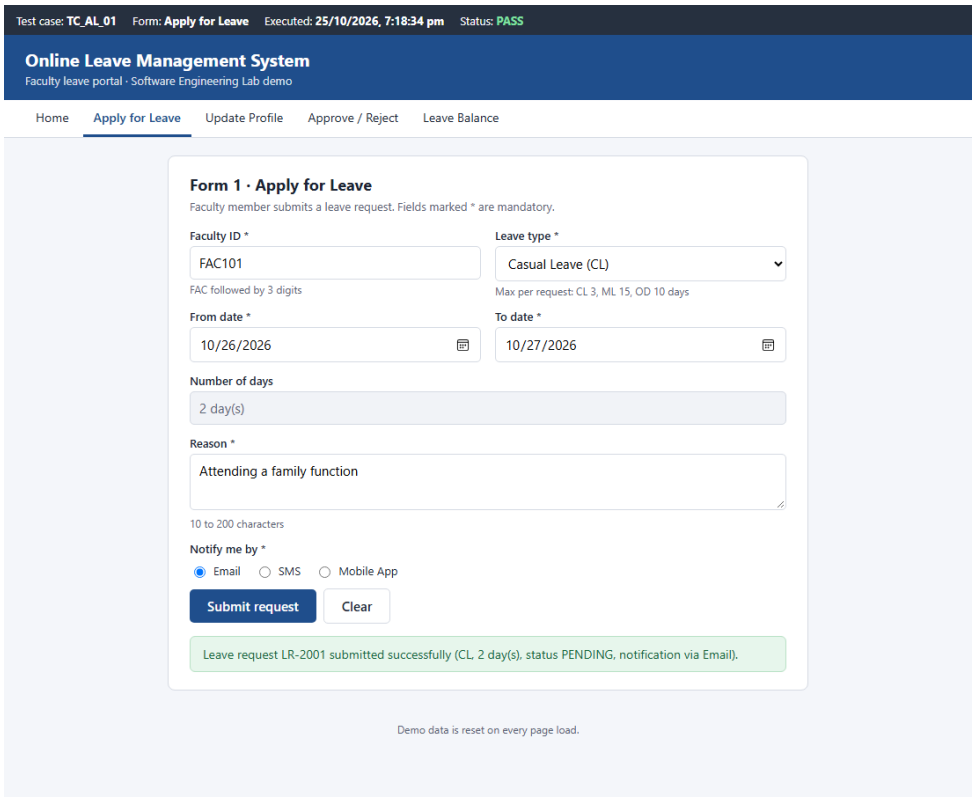


Figure 1: TC_AL_01 - Form 1 - valid Casual Leave request accepted

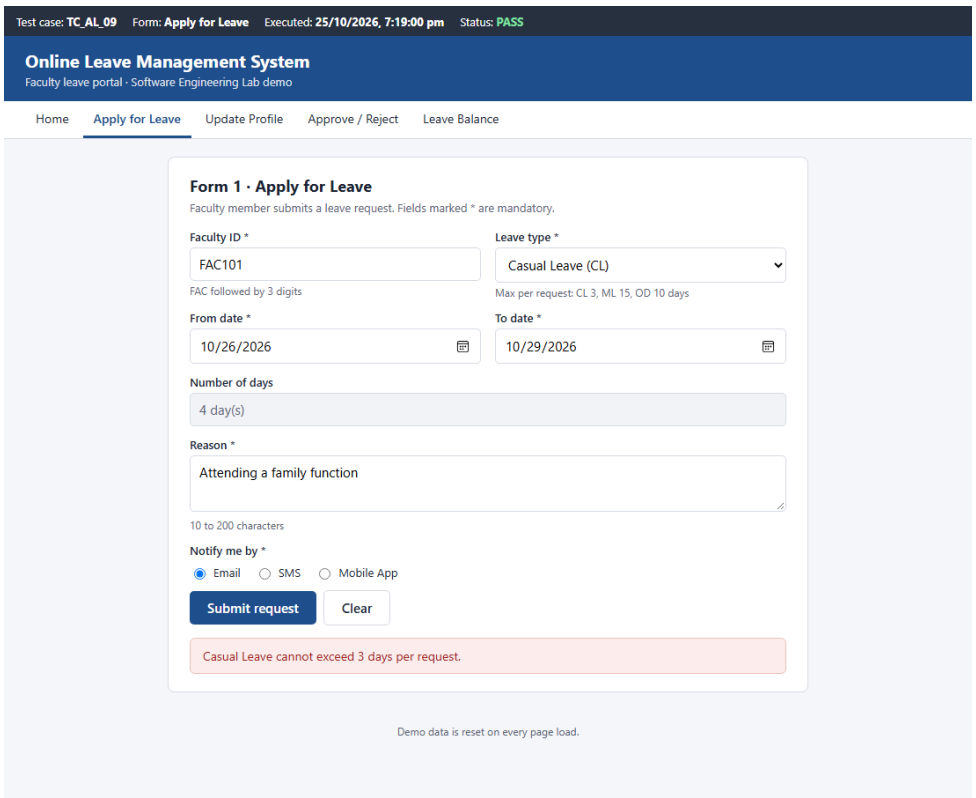


Figure 2: TC_AL_09 - Form 1 - Casual Leave of 4 days rejected (BVA, max + 1)

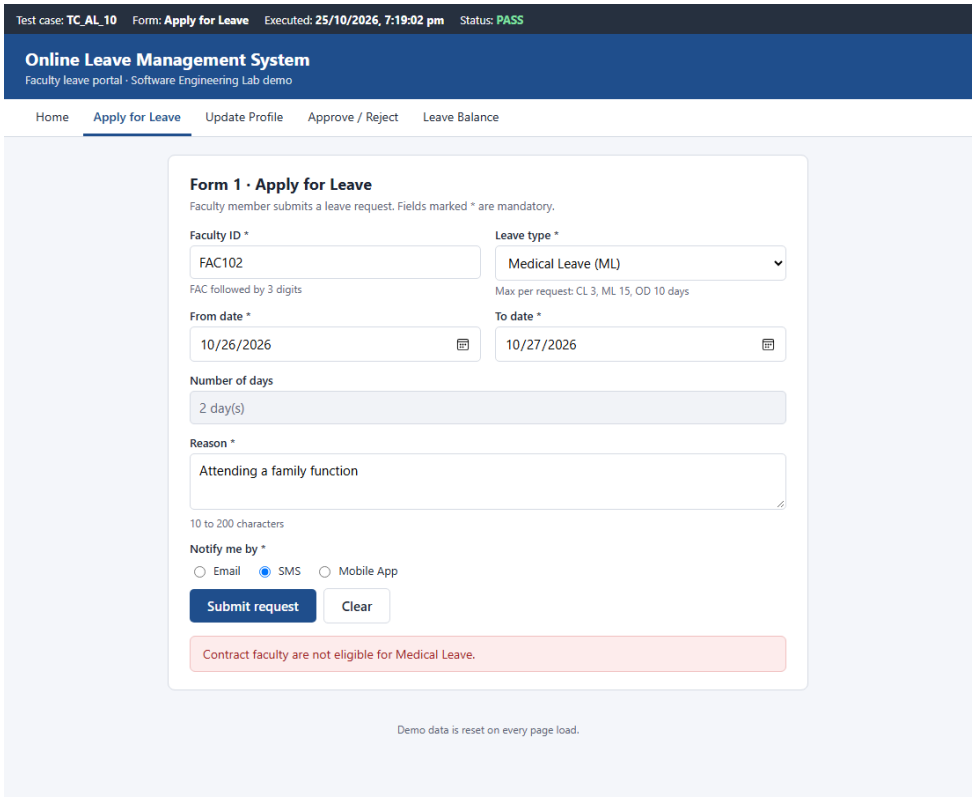


Figure 3: TC_AL_10 - Form 1 - contract faculty not eligible for Medical Leave

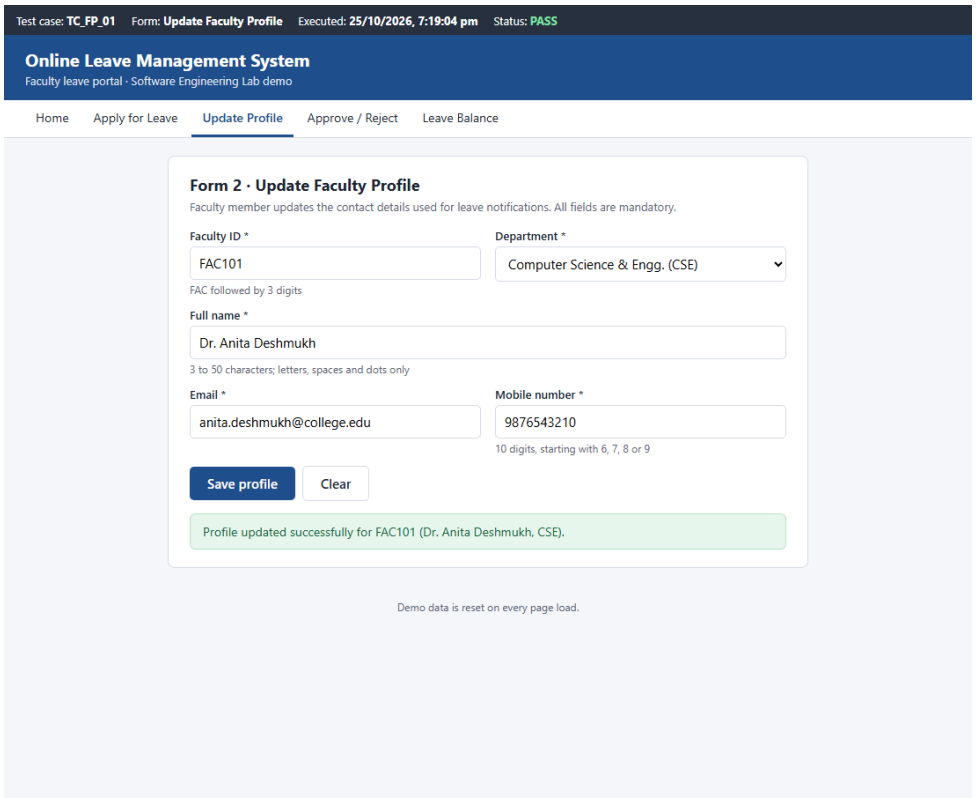


Figure 4: TC_FP_01 - Form 2 - valid profile saved

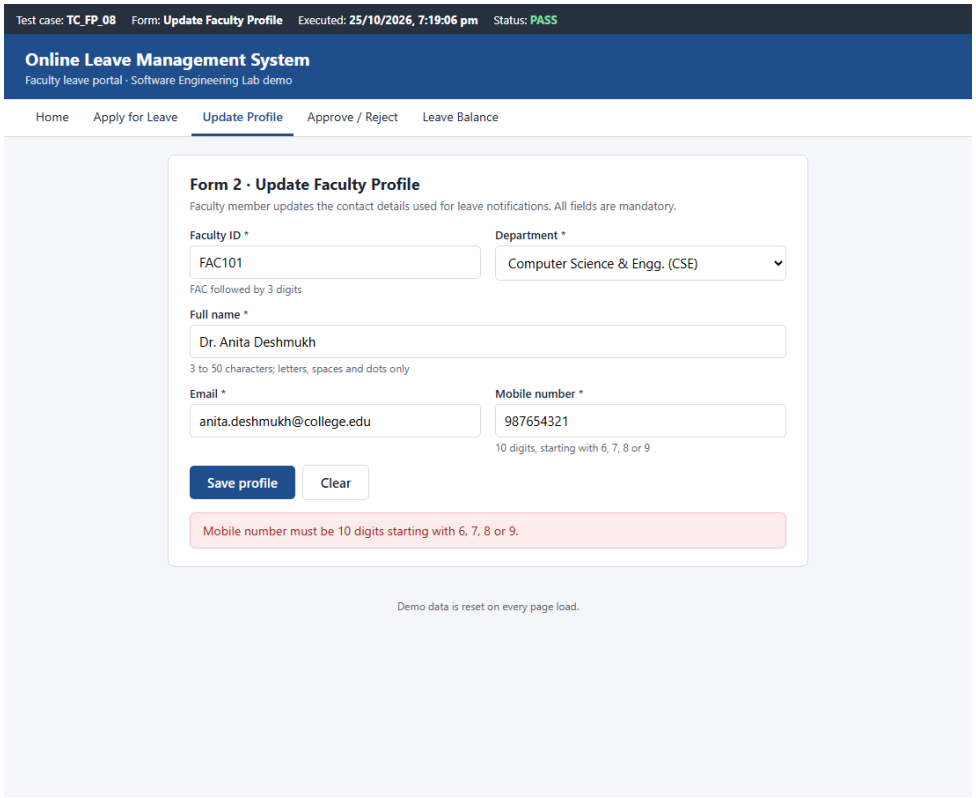


Figure 5: TC_FP_08 - Form 2 - 9-digit mobile number rejected (BVA)

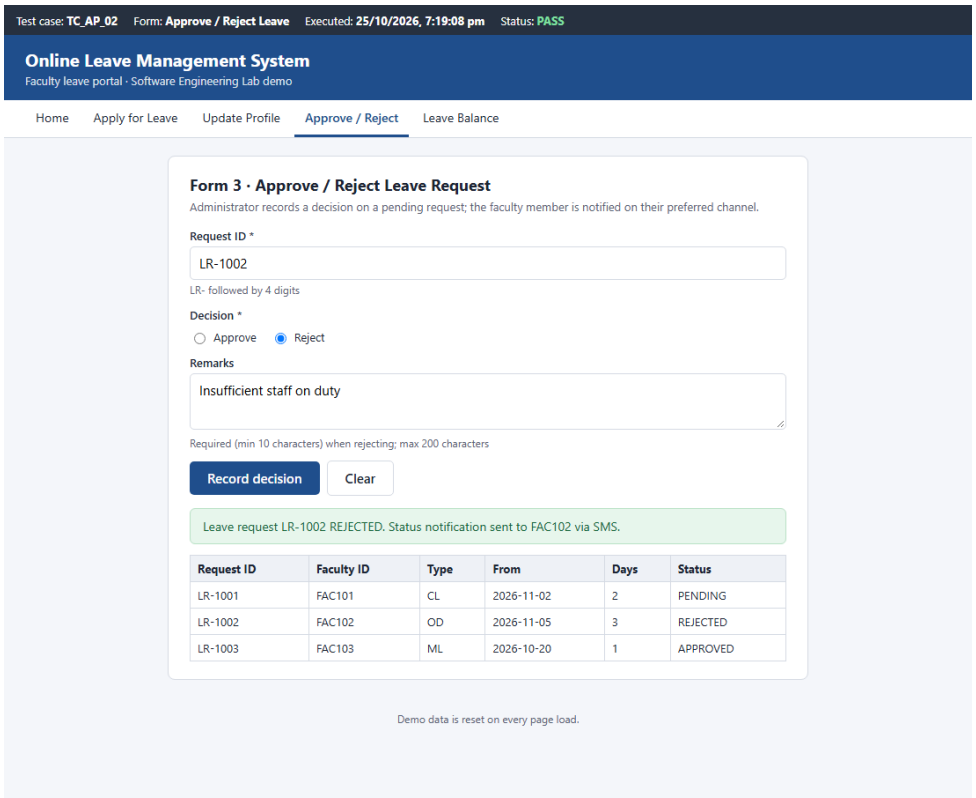


Figure 6: TC_AP_02 - Form 3 - pending request rejected with remarks

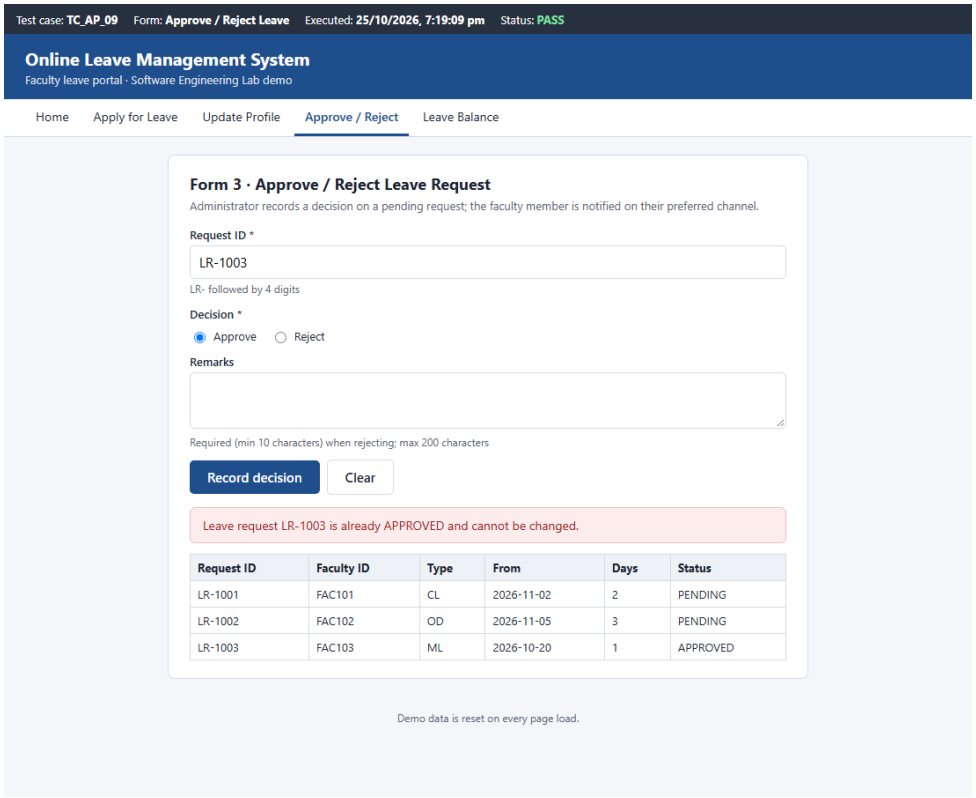


Figure 7: TC_AP_09 - Form 3 - already approved request cannot be changed

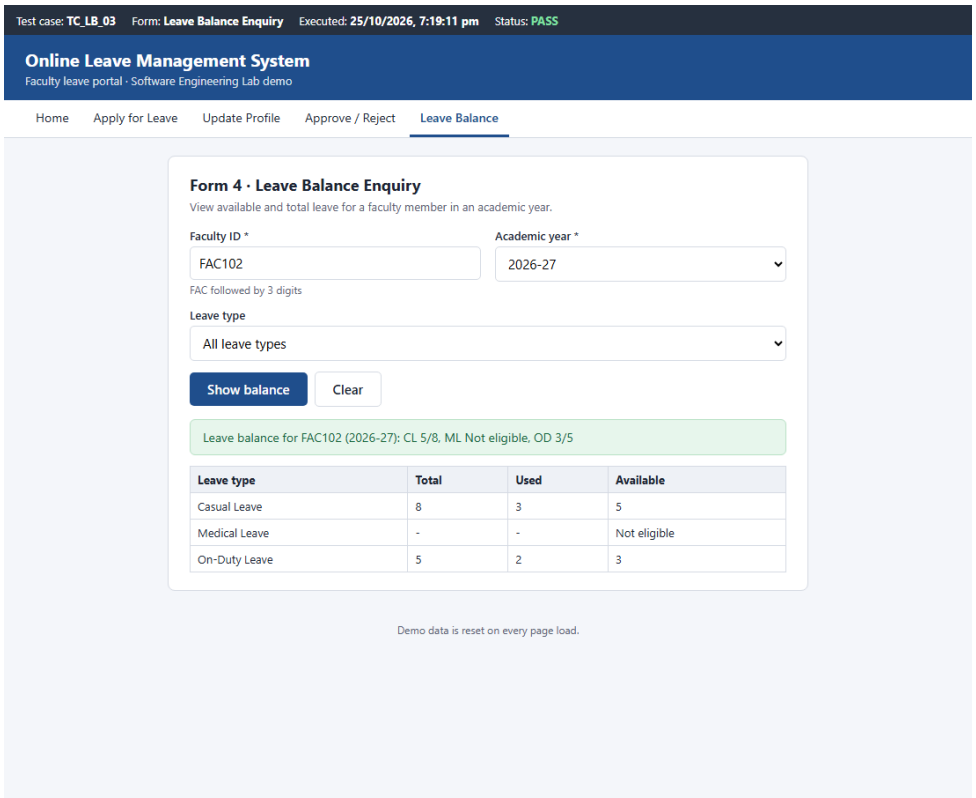


Figure 8: TC_LB_03 - Form 4 - balance of contract faculty

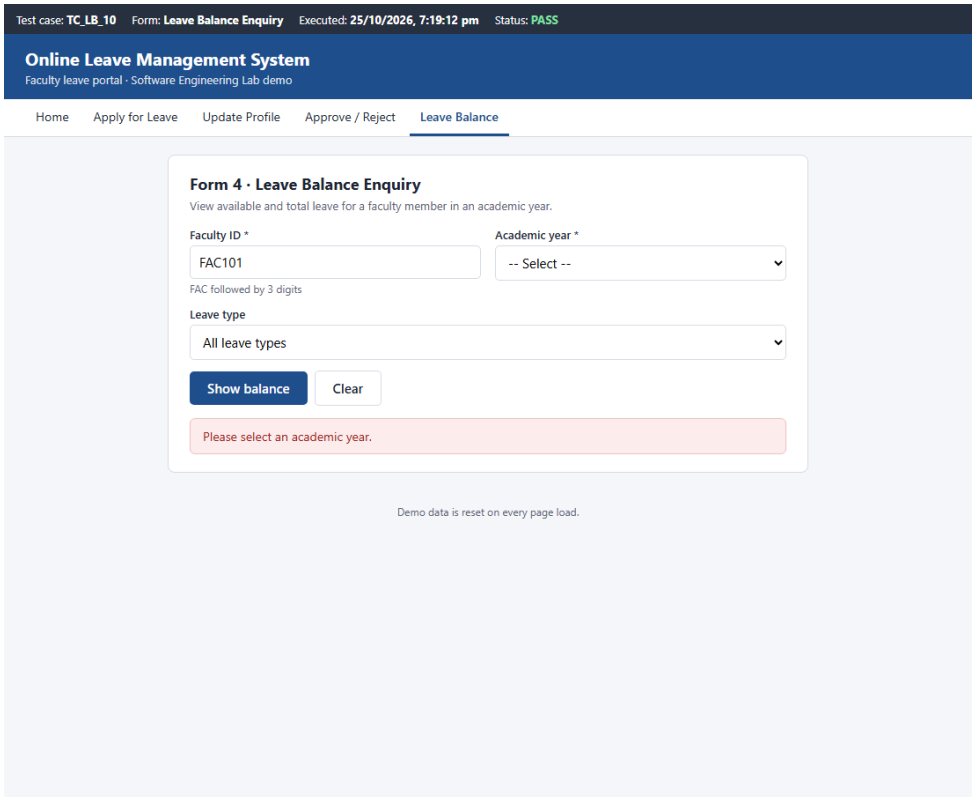


Figure 9: TC_LB_10 - Form 4 - academic year not selected

Expt-8 · Black-box test run · Online Leave Management System

49 test cases executed · 49 passed · 0 failed · Sun Oct 25 2026 19:18:26 GMT+0530 (India Standard Time)

TC ID	Form	Scenario	Expected output	Actual output	Status
TC_AL_01	Apply for Leave	Valid Casual Leave request of 2 days	Leave request LR-2001 submitted successfully (CL, 2 day(s), status PENDING, notification via Email).	Leave request LR-2001 submitted successfully (CL, 2 day(s), status PENDING, notification via Email).	PASS
TC_AL_02	Apply for Leave	Faculty ID left blank	Faculty ID is required.	Faculty ID is required.	PASS
TC_AL_03	Apply for Leave	Faculty ID in wrong format (only 2 digits)	Faculty ID must be FAC followed by 3 digits (e.g. FAC101).	Faculty ID must be FAC followed by 3 digits (e.g. FAC101).	PASS
TC_AL_04	Apply for Leave	Faculty ID in correct format but not registered	Faculty ID FAC999 is not registered.	Faculty ID FAC999 is not registered.	PASS
TC_AL_05	Apply for Leave	From date one day in the past	From date cannot be in the past.	From date cannot be in the past.	PASS
TC_AL_06	Apply for Leave	Leave for today only (From = To = today)	Leave request LR-2001 submitted successfully (CL, 1 day(s), status PENDING, notification via Email).	Leave request LR-2001 submitted successfully (CL, 1 day(s), status PENDING, notification via Email).	PASS
TC_AL_07	Apply for Leave	To date earlier than From date	To date cannot be before From date.	To date cannot be before From date.	PASS
TC_AL_08	Apply for Leave	Casual Leave of exactly 3 days (upper limit)	Leave request LR-2001 submitted successfully (CL, 3 day(s), status PENDING, notification via Email).	Leave request LR-2001 submitted successfully (CL, 3 day(s), status PENDING, notification via Email).	PASS
TC_AL_09	Apply for Leave	Casual Leave of 4 days (above limit)	Casual Leave cannot exceed 3 days per request.	Casual Leave cannot exceed 3 days per request.	PASS
TC_AL_10	Apply for Leave	Contract faculty applies for Medical Leave	Contract faculty are not eligible for Medical Leave.	Contract faculty are not eligible for Medical Leave.	PASS
TC_AL_11	Apply for Leave	Medical Leave of 3 days with only 2 days balance	Insufficient ML balance (available: 2 day(s)).	Insufficient ML balance (available: 2 day(s)).	PASS
TC_AL_12	Apply for Leave	Reason of 9 characters	Reason must be 10 to 200 characters.	Reason must be 10 to 200 characters.	PASS
TC_AL_13	Apply for Leave	Reason of exactly 10 characters	Leave request LR-2001 submitted successfully (CL, 2 day(s), status PENDING, notification via Email).	Leave request LR-2001 submitted successfully (CL, 2 day(s), status PENDING, notification via Email).	PASS
TC_AL_14	Apply for Leave	No notification channel selected	Please select a notification channel.	Please select a notification channel.	PASS
TC_AL_15	Apply for Leave	Valid On-Duty Leave of 5 days, Mobile App notification	Leave request LR-2001 submitted successfully (OD, 5 day(s), status PENDING, notification via Mobile App).	Leave request LR-2001 submitted successfully (OD, 5 day(s), status PENDING, notification via Mobile App).	PASS
TC_FP_01	Update Faculty Profile	All profile fields valid	Profile updated successfully for FAC101 (Dr. Anita Deshmukh, CSE).	Profile updated successfully for FAC101 (Dr. Anita Deshmukh, CSE).	PASS
TC_FP_02	Update Faculty Profile	Name contains digits	Name must be 3 to 50 characters (letters, spaces and dots only).	Name must be 3 to 50 characters (letters, spaces and dots only).	PASS
TC_FP_03	Update Faculty Profile	Name of 2 characters	Name must be 3 to 50 characters (letters, spaces and dots only).	Name must be 3 to 50 characters (letters, spaces and dots only).	PASS
TC_FP_04	Update Faculty Profile	Name of exactly 3 characters	Profile updated successfully for FAC101 (Ali, CSE).	Profile updated successfully for FAC101 (Ali, CSE).	PASS
TC_FP_05	Update Faculty Profile	Name of exactly 50 characters	Profile updated successfully for FAC101 (Dr. Anita Deshmukh Ramchandra Kulkarni Patil Joshi, CSE).	Profile updated successfully for FAC101 (Dr. Anita Deshmukh Ramchandra Kulkarni Patil Joshi, CSE).	PASS
TC_FP_06	Update Faculty Profile	Name of 51 characters	Name must be 3 to 50 characters (letters, spaces and dots only).	Name must be 3 to 50 characters (letters, spaces and dots only).	PASS
TC_FP_07	Update Faculty Profile	Email without @ symbol	Enter a valid email address (e.g. name@college.edu).	Enter a valid email address (e.g. name@college.edu).	PASS
TC_FP_08	Update Faculty Profile	Mobile number of 9 digits	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	PASS
TC_FP_09	Update Faculty Profile	Mobile number of 11 digits	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	PASS
TC_FP_10	Update Faculty Profile	Mobile number starting with 5	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	Mobile number must be 10 digits starting with 6, 7, 8 or 9.	PASS
TC_FP_11	Update Faculty Profile	Department not selected	Please select a department.	Please select a department.	PASS
TC_FP_12	Update Faculty Profile	Faculty ID not registered	Faculty ID FAC999 is not registered.	Faculty ID FAC999 is not registered.	PASS
TC_AP_01	Approve / Reject Leave	Approve a pending request without remarks	Leave request LR-1001 APPROVED. Status notification sent to FAC101 via Email.	Leave request LR-1001 APPROVED. Status notification sent to FAC101 via Email.	PASS
TC_AP_02	Approve / Reject Leave	Reject a pending request with remarks	Leave request LR-1002 REJECTED. Status notification sent to FAC102 via SMS.	Leave request LR-1002 REJECTED. Status notification sent to FAC102 via SMS.	PASS
TC_AP_03	Approve / Reject Leave	Reject without remarks	Remarks of at least 10 characters are required when rejecting.	Remarks of at least 10 characters are required when rejecting.	PASS
TC_AP_04	Approve / Reject Leave	Reject with 9-character remarks	Remarks of at least 10 characters are required when rejecting.	Remarks of at least 10 characters are required when rejecting.	PASS
TC_AP_05	Approve / Reject Leave	Reject with exactly 10-character remarks	Leave request LR-1002 REJECTED. Status notification sent to FAC102 via SMS.	Leave request LR-1002 REJECTED. Status notification sent to FAC102 via SMS.	PASS
TC_AP_06	Approve / Reject Leave	Request ID left blank	Request ID is required.	Request ID is required.	PASS
TC_AP_07	Approve / Reject Leave	Request ID without LR- prefix	Request ID must be LR- followed by 4 digits (e.g. LR-1001).	Request ID must be LR- followed by 4 digits (e.g. LR-1001).	PASS
TC_AP_08	Approve / Reject Leave	Request ID that does not exist	Leave request LR-9999 not found.	Leave request LR-9999 not found.	PASS
TC_AP_09	Approve / Reject Leave	Request already approved earlier	Leave request LR-1003 is already APPROVED and cannot be changed.	Leave request LR-1003 is already APPROVED and cannot be changed.	PASS
TC_AP_10	Approve / Reject Leave	Decision not selected	Please select Approve or Reject.	Please select Approve or Reject.	PASS
TC_AP_11	Approve / Reject Leave	Approve with remarks of exactly 200 characters	Leave request LR-1001 APPROVED. Status notification sent to FAC101 via Email.	Leave request LR-1001 APPROVED. Status notification sent to FAC101 via Email.	PASS
TC_AP_12	Approve / Reject Leave	Approve with remarks of 201 characters	Remarks cannot exceed 200 characters.	Remarks cannot exceed 200 characters.	PASS
TC_LB_01	Leave Balance Enquiry	Permanent faculty, all leave types	Leave balance for FAC101 (2026-27): CL 8/12, ML 10/10, OD 5/7	Leave balance for FAC101 (2026-27): CL 8/12, ML 10/10, OD 5/7	PASS
TC_LB_02	Leave Balance Enquiry	Permanent faculty, Casual Leave only	Leave balance for FAC101 (2026-27): CL 8/12	Leave balance for FAC101 (2026-27): CL 8/12	PASS
TC_LB_03	Leave Balance Enquiry	Contract faculty, all leave types (ML not eligible)	Leave balance for FAC102 (2026-27): CL 5/8, ML Not eligible, OD 3/5	Leave balance for FAC102 (2026-27): CL 5/8, ML Not eligible, OD 3/5	PASS
TC_LB_04	Leave Balance Enquiry	Permanent faculty, Medical Leave only	Leave balance for FAC103 (2026-27): ML 2/10	Leave balance for FAC103 (2026-27): ML 2/10	PASS
TC_LB_05	Leave Balance Enquiry	Academic year with no records	No leave records found for FAC101 in 2025-26.	No leave records found for FAC101 in 2025-26.	PASS
TC_LB_06	Leave Balance Enquiry	Faculty ID in lower case with spaces	Leave balance for FAC101 (2026-27): CL 8/12, ML 10/10, OD 5/7	Leave balance for FAC101 (2026-27): CL 8/12, ML 10/10, OD 5/7	PASS
TC_LB_07	Leave Balance Enquiry	Faculty ID left blank	Faculty ID is required.	Faculty ID is required.	PASS
TC_LB_08	Leave Balance Enquiry	Faculty ID with letter O instead of zero	Faculty ID must be FAC followed by 3 digits (e.g. FAC101).	Faculty ID must be FAC followed by 3 digits (e.g. FAC101).	PASS
TC_LB_09	Leave Balance Enquiry	Faculty ID not registered	Faculty ID FAC999 is not registered.	Faculty ID FAC999 is not registered.	PASS
TC_LB_10	Leave Balance Enquiry	Academic year not selected	Please select an academic year.	Please select an academic year.	PASS

Figure 10: Complete test run - 49 test cases executed, 49 passed, 0 failed

Conclusion

A test case document was prepared for four form interfaces of the Online Leave Management System website: Apply for Leave, Update Faculty Profile, Approve / Reject Leave and Leave Balance Enquiry. Using Equivalence Partitioning and Boundary Value Analysis, 49 test cases were designed with their inputs and expected outputs. All test cases were executed through the user interface and the actual outputs were recorded: 49 passed and 0 failed.